

ASEN 5090
Fall 2026
HW 1
Jash Bhalavat

Problem 1

HW #1

Problem 1] a) $e_n^{PL1} = 550 \text{ m} = R_n^{PL1} + c d_n \rightarrow (1)$

ME Book $e_n^{PL2} = 500 \text{ m} = R_n^{PL2} + c d_n$

1-4 $= 1000 - R_n^{PL1} + c d_n$

$-500 = -R_n^{PL1} + c d_n \rightarrow (2)$

Add (1), (2) $50 = 2c d_n \rightarrow d_n = 8.3391 \times 10^{-8} \text{ s}$ ($c = 2.99792458 \times 10^8 \text{ m/s}$)

$R_n^{PL1} = 550 - c d_n = 525 \text{ m} = R_n$

$R_n^{PL2} = 1000 - R_n^{PL1} = 475 \text{ m} = R_n^{PL2}$

b) $e_n^{PL1} = 400 \text{ m} = R_n^{PL1} + c d_n \rightarrow (1)$

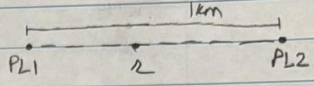
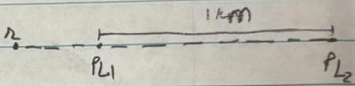
$e_n^{PL2} = 1400 \text{ m} = R_n^{PL2} + c d_n = 1000 - R_n^{PL1} + c d_n$

$400 = -R_n^{PL1} + c d_n \rightarrow (2)$

Add (1), (2) $0 = 2c d_n \rightarrow d_n = 0 \text{ sec}$

$R_n^{PL1} = 400 \text{ m}$

$R_n^{PL2} = 1400 \text{ m}$

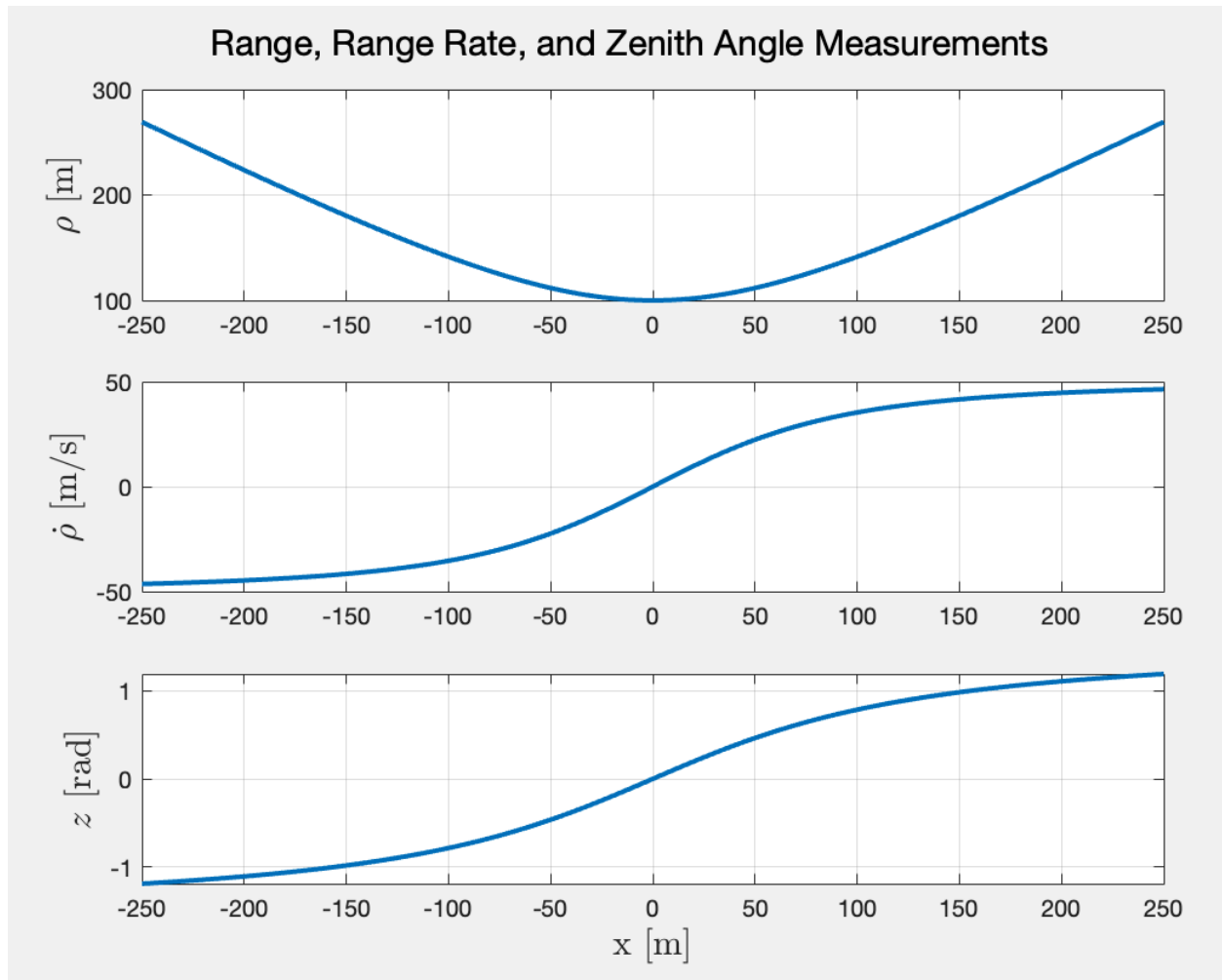
Problem 2a

Problem 2] a) Range: $r = \sqrt{h_0^2 + x_A^2}$

Range rate: $\dot{r} = \frac{1}{2} (h_0^2 + x_A^2)^{-1/2} \cdot (2x_A \dot{x}_A) = \frac{x_A \dot{x}_A}{\sqrt{h_0^2 + x_A^2}} = \frac{x_A \dot{x}_A}{r}$

Zenith angle: $z = \tan^{-1} \left(\frac{x_A}{h_0} \right)$

Problem 2b



```

12     n = 500;
13
14 %% Problem 2
15
16 v0 = 50; % m/s
17 h0 = 100; % m
18 x0 = 250; % m
19
20 xA = linspace(-x0, x0, n);
21
22 for i = 1:n
23     rho(i) = sqrt(h0^2 + xA(i)^2); % m
24     rho_dot(i) = v0*xA(i) / rho(i); % m/s
25     z(i) = atan2(xA(i), h0);
26 end
27
28 % Plot
29 figure(1)
30 subplot(3,1,1)
31 plot(xA, rho, 'LineWidth', 2)
32 ylabel('$\rho$ [m]', 'Interpreter', 'Latex', 'FontSize', 15)
33 grid on
34
35 subplot(3,1,2)
36 plot(xA, rho_dot, 'LineWidth', 2)
37 ylabel('$\dot{\rho}$ [m/s]', 'Interpreter', 'Latex', 'FontSize', 15)
38 grid on
39
40 subplot(3,1,3)
41 plot(xA, z, 'LineWidth', 2)
42 ylabel('$z$ [rad]', 'Interpreter', 'Latex', 'FontSize', 15)
43 xlabel('x [m]', 'Interpreter', 'Latex', 'FontSize', 15)
44 grid on
45
46 sgtitle("Range, Range Rate, and Zenith Angle Measurements")
47
48

```

The full set of code is published in the Appendix after Problem 4.

Problem 2c

Utility:

As seen above, range cannot provide direction, only the magnitude of aircraft horizontal position. Range rate and zenith angle can provide both direction and magnitude.

Other mathematical constraints make some of the equations less useful - If h_0 is zero, zenith angle can no longer be calculated. Similarly, if range is 0, range rate cannot be calculated.

$x_A = 0$

Range: $\frac{\partial e}{\partial x_A} = \frac{2x_A}{2\sqrt{h_0^2 + x_A^2}} = \frac{x_A}{e}$

Range rate: $\frac{\partial e}{\partial x_A} = \sqrt{h_0^2 + x_A^2} \cdot V_0 - x_A V_0 \cdot \frac{1}{2}(h_0^2 + x_A^2)^{-1/2} \cdot 2x_A = \frac{eV_0 - V_0 x_A^2}{e}$

$\frac{V_0}{e} - \frac{V_0 x_A^2}{e^3} = \frac{V_0 e^2 - V_0 x_A^2}{e^3} = \frac{V_0 (h_0^2 + x_A^2) - V_0 x_A^2}{e^3} = \frac{V_0 h_0^2}{e^3}$

Zenith angle: $\frac{\partial e}{\partial x_A} = \frac{1}{1 + \frac{x_A^2}{h_0^2}} = \frac{h_0^2}{h_0^2 + x_A^2} = \frac{h_0^2}{e^2}$

Sensitivity:

Range measurement: this measurement is more sensitive when the aircraft's horizontal position is further away from the ground station as sensitivity is 0 when aircraft position is 0 (as seen in the plot above, the range slope decreases closer to 0).

Range rate measurement: this measurement is more sensitive when the aircraft's horizontal position is closer to the ground station (as seen in the plot above, the range rate slope increases closer to 0). This is mathematically evident in the fact that the aircraft's horizontal position is seen only in the denominator, so if the denominator is very small (0), the overall value (sensitivity) is high.

Zenith angle measurement: this measurement is also more sensitive when the aircraft's horizontal position is closer to the ground station (as seen in the plot above, the zenith angle measurement slope changes more dramatically closer to 0). Similar to range rate, this can be mathematically evident in the fact that the aircraft's horizontal position is seen only in the denominator.

Problem 2d

Using the same equations as part (c) to see sensitivity of these measurements when h_0 decreases:

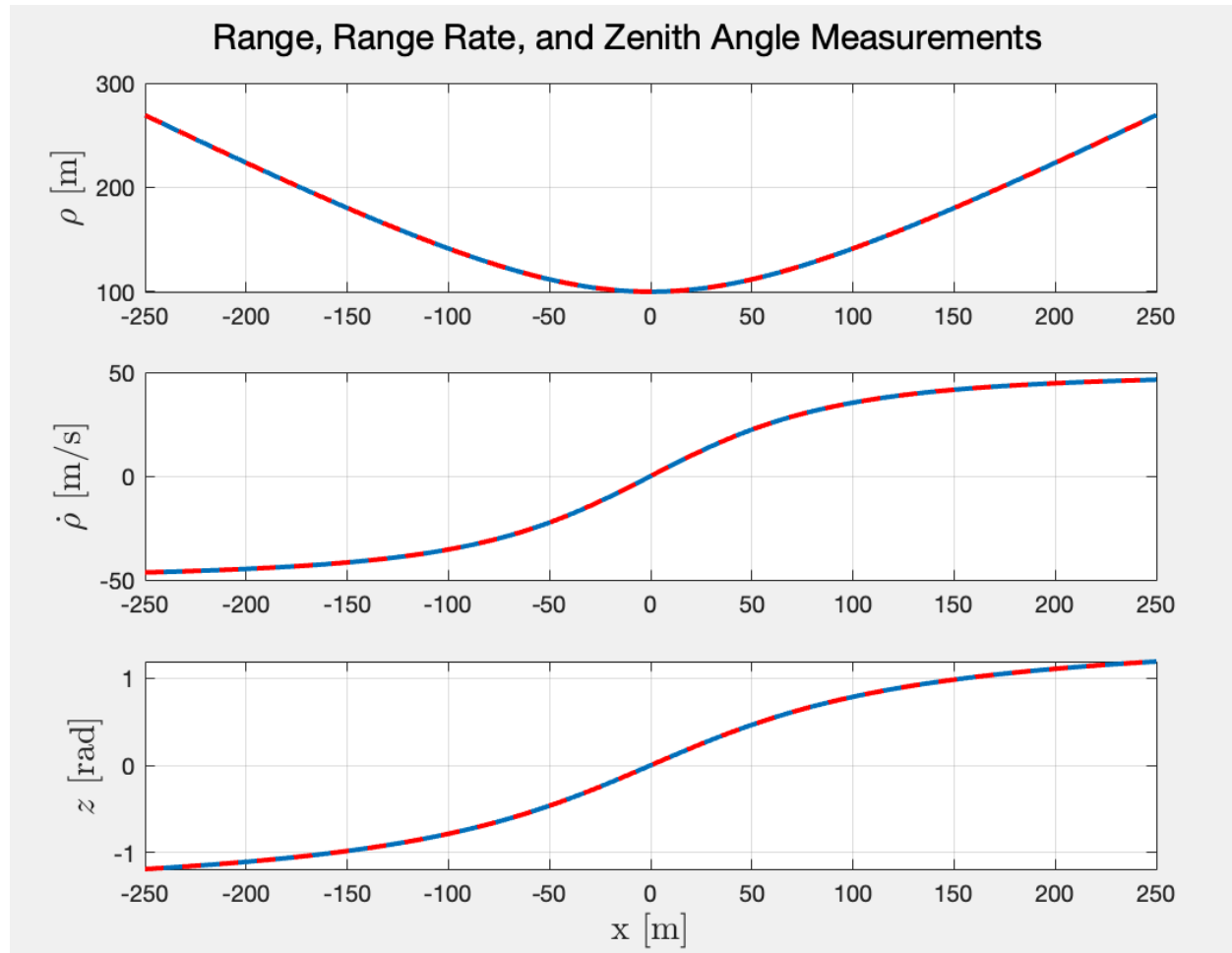
Range measurement: since h_0 is in the denominator, as h_0 decreases the sensitivity gets better.

Range rate measurement: when aircraft's horizontal position is 0, the sensitivity is $\frac{v_0}{h_0}$ and if h_0 decreases, the sensitivity gets better.

Zenith angle measurement: when aircraft's horizontal position is 0, the sensitivity is $\frac{1}{h_0}$ and if h_0 decreases, the sensitivity gets better.

Problem 2e

If there is an error in the assumed height, all three measurements will have induced errors. The magnitude of error induced in the measurements will vary. A plot is created with 10% change in h_0 (red dotted lines indicate measurements with errors):



and as seen, the error bands follow the measurements pretty close. Sensitivity would have a higher effect.

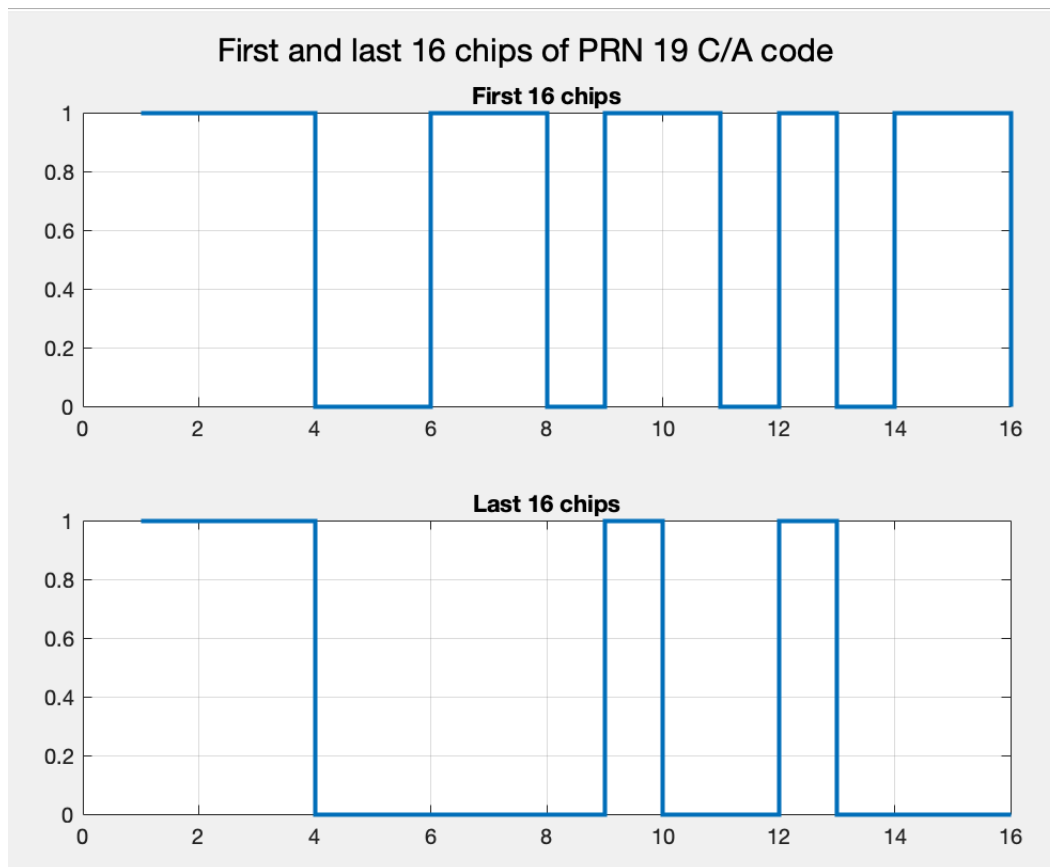
Range - height is not used directly, but instead assumed through the range measurement, so this won't affect range as much.

Range rate and zenith - both use height directly, so an error in the assumed height would have a big impact on these measurements.

Problem 3a

Using [IS-GPS-200D](#), it was found that for PRN 19, S1 and S2 are 3 and 6 respectively.

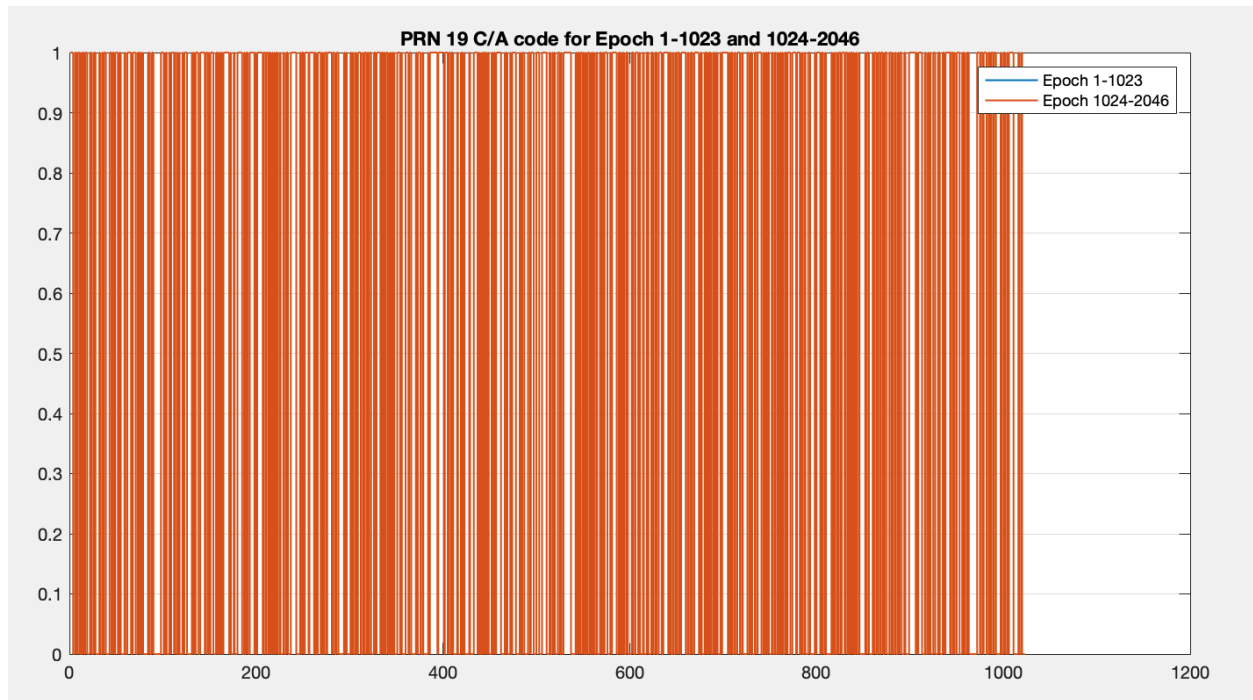
Using the algorithm provided in the problem statement, a 1023-chip C/A code for PRN 19 was generated. The first 16 and last 16 chips of the code are shown below:



When expressed in hex, the first 16 chips are **0xE6D6** and the last 16 chips are **0xE090**.

Problem 3b

Repeating the C/A code for epoch 1024-2046:

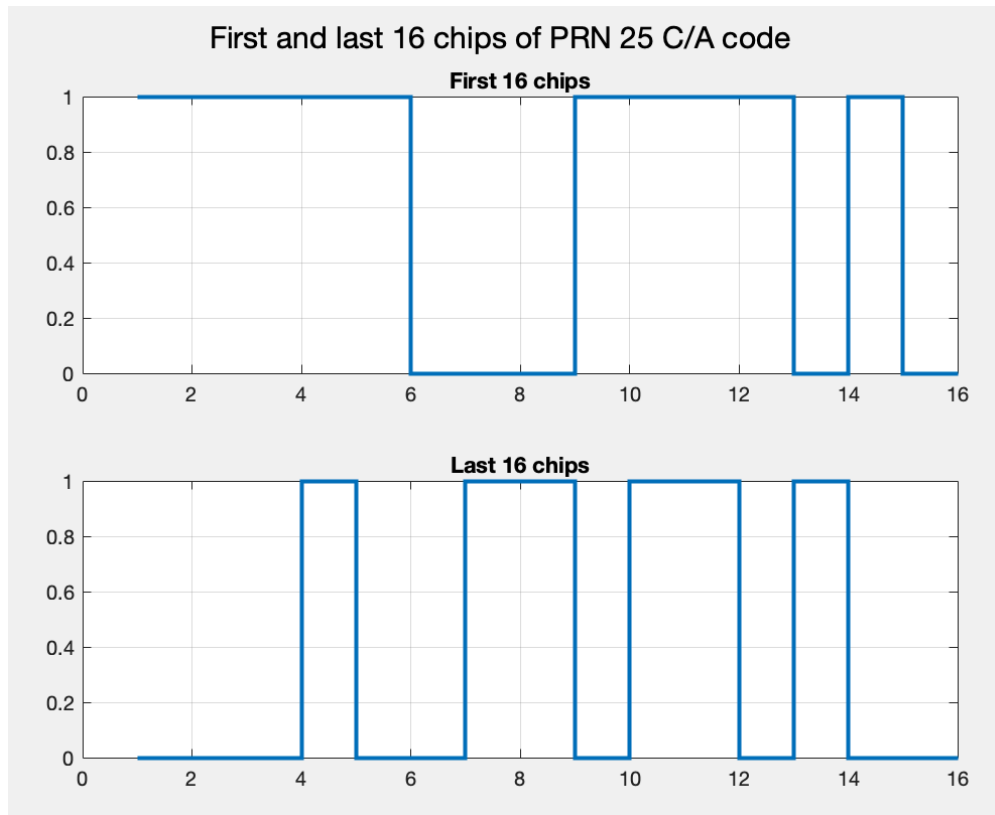


The 1023-element vectors are identical. This suggests that this C/A code repeats itself every 1023 cycles.

Problem 3c

Using [IS-GPS-200D](#), it was found that for PRN 25, S1 and S2 are 5 and 7 respectively.

Using the algorithm provided in the problem statement, a 1023-chip C/A code for PRN 25 was generated. The first 16 and last 16 chips of the code are shown below:

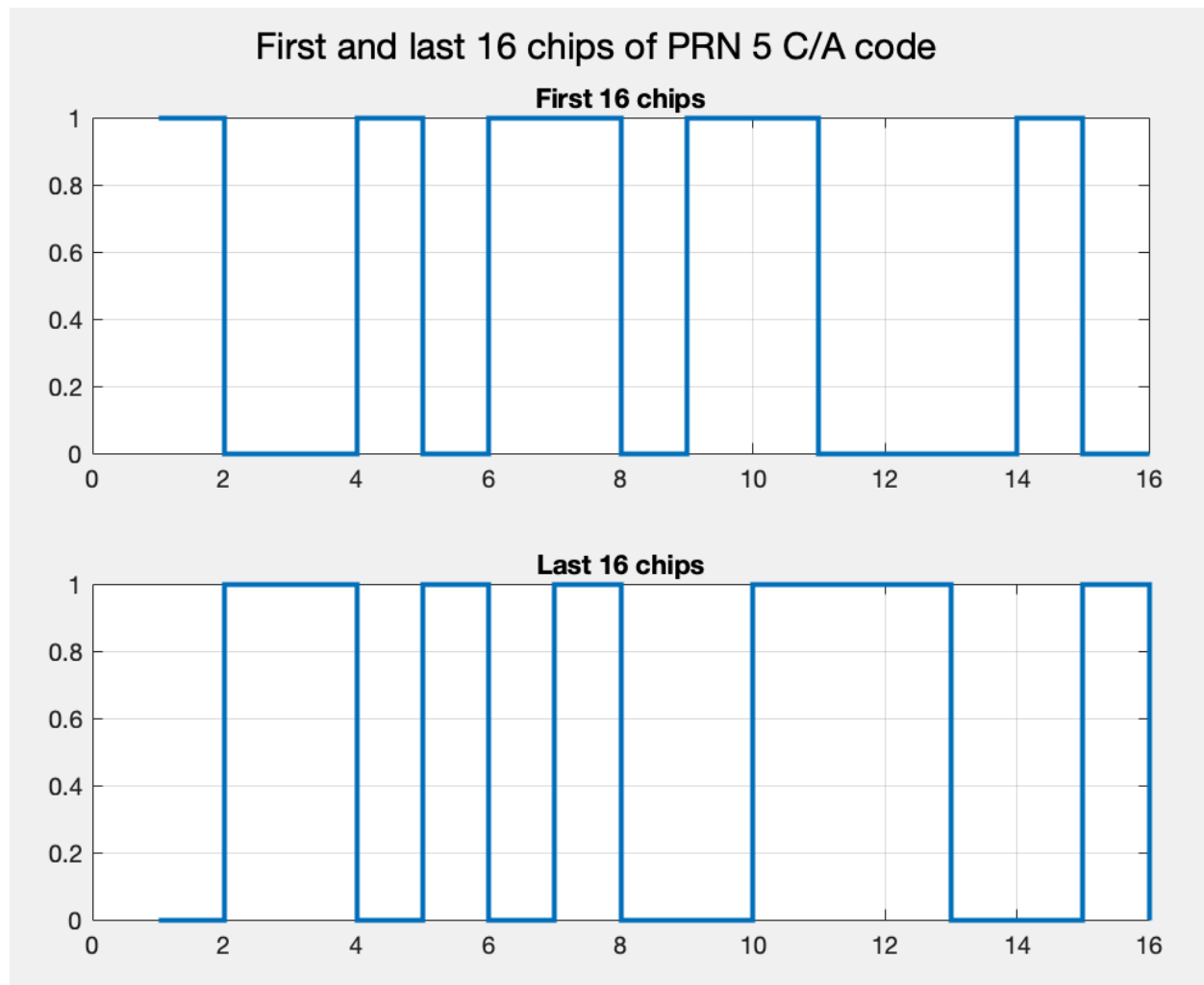


When expressed in hex, the first 16 chips are **0xF8F4** and the last 16 chips are **0x1368**.

Problem 3d

Using [IS-GPS-200D](#), it was found that for PRN 5, S1 and S2 are 1 and 9 respectively.

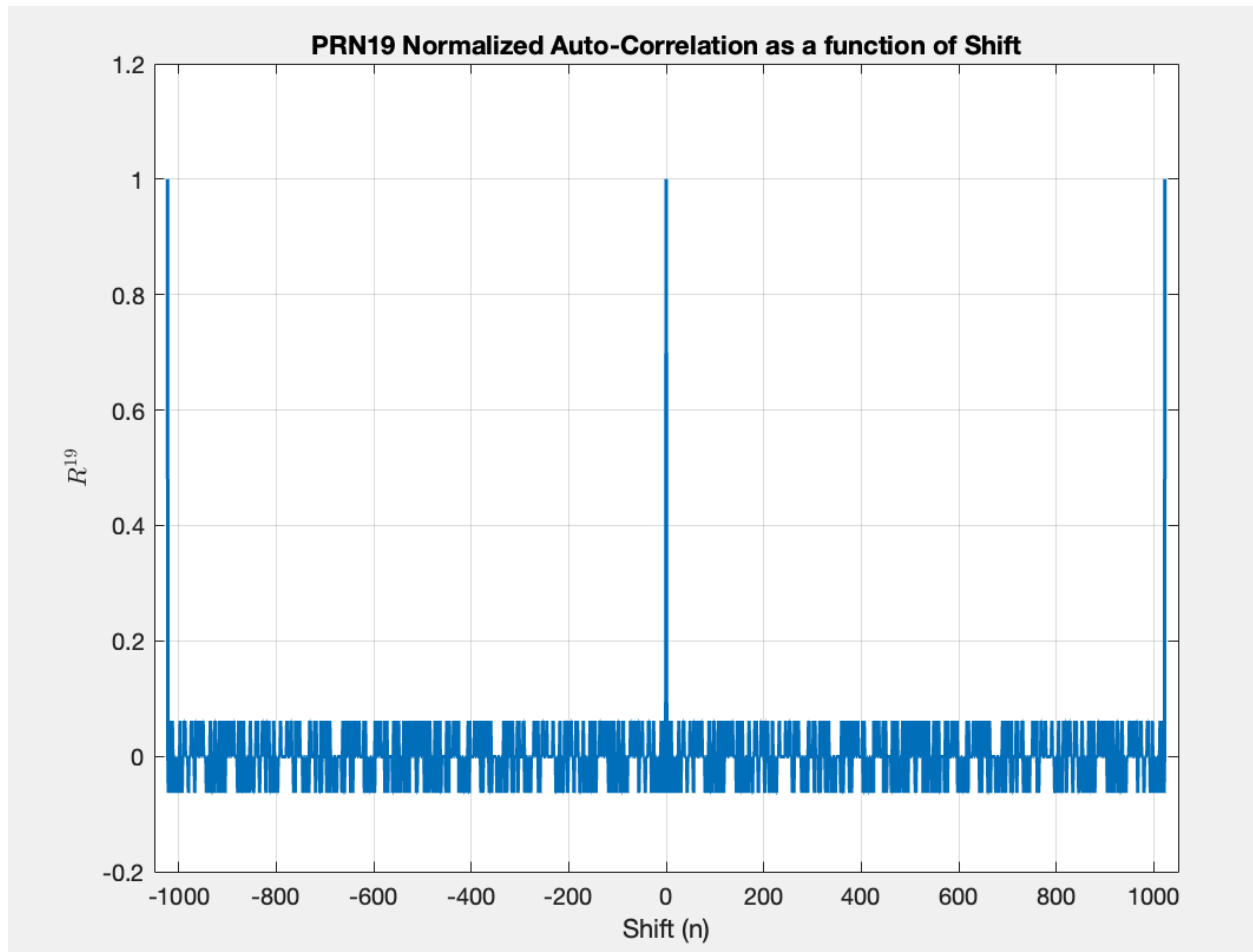
Using the algorithm provided in the problem statement, a 1023-chip C/A code for PRN 5 was generated. The first 16 and last 16 chips of the code are shown below:



When expressed in hex, the first 16 chips are **0x96C4** and the last 16 chips are **0x6A72**.

Problem 4a

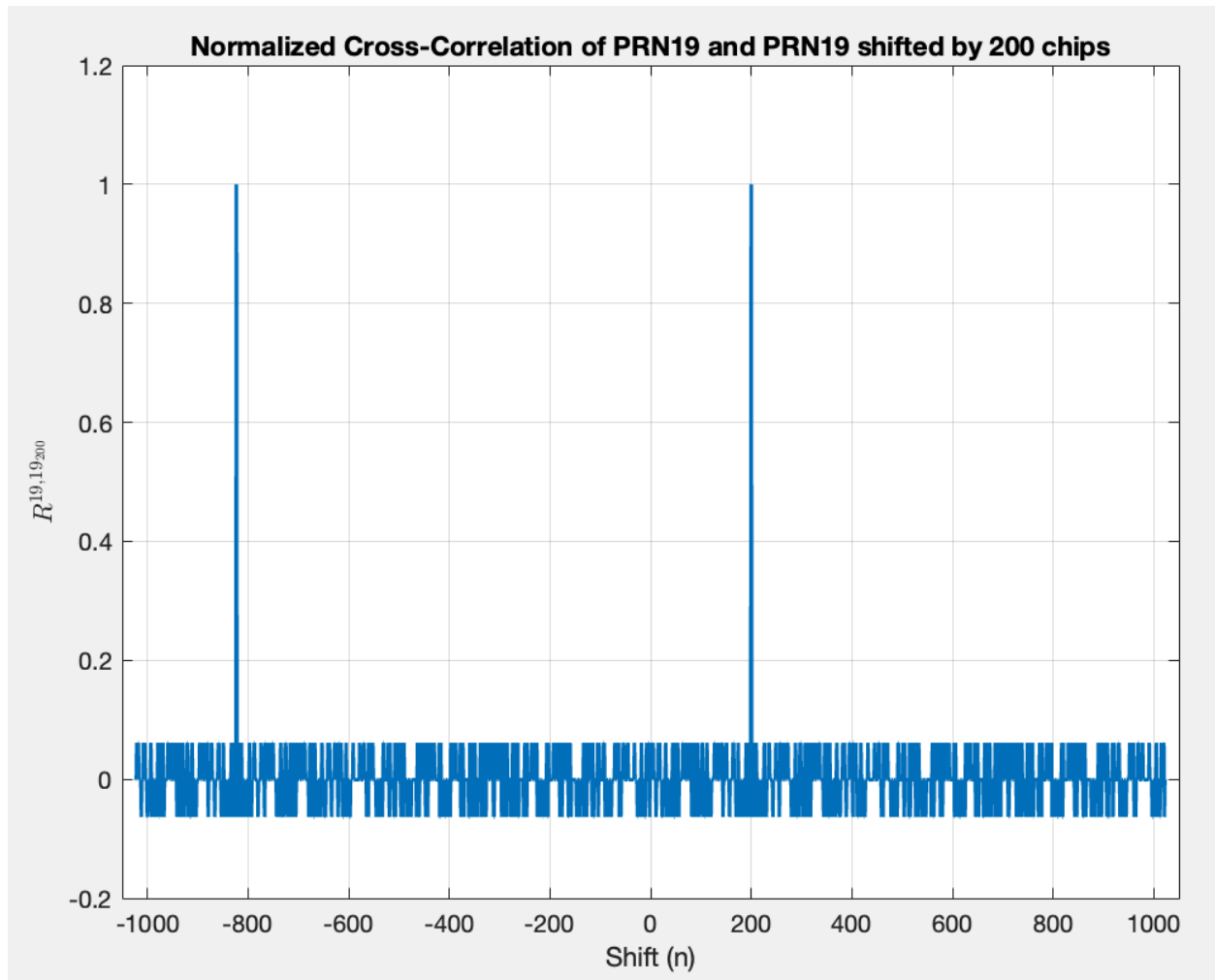
Here's the auto-correlation function of PRN 19



Maximum auto-correlation happens where expected - $n = -1023, 0, 1023$

Problem 4b

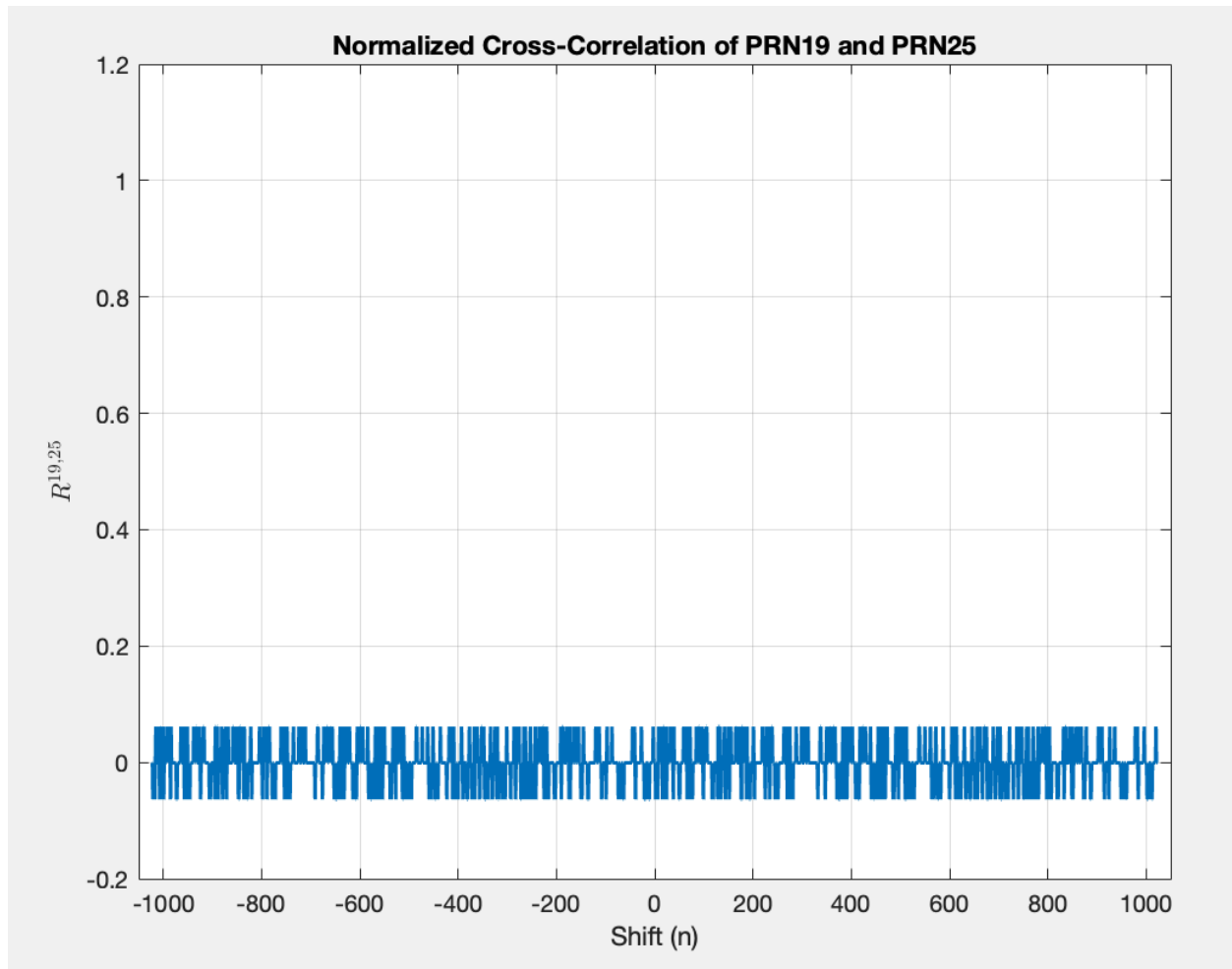
Here's the cross-correlation function for PRN19 and PRN19 delayed by 200 chips:



The peaks show up at $n = 200$ and $n = -823$. I expected a peak at 200 which would have meant the peaks shifted to the right by 200.

Problem 4c

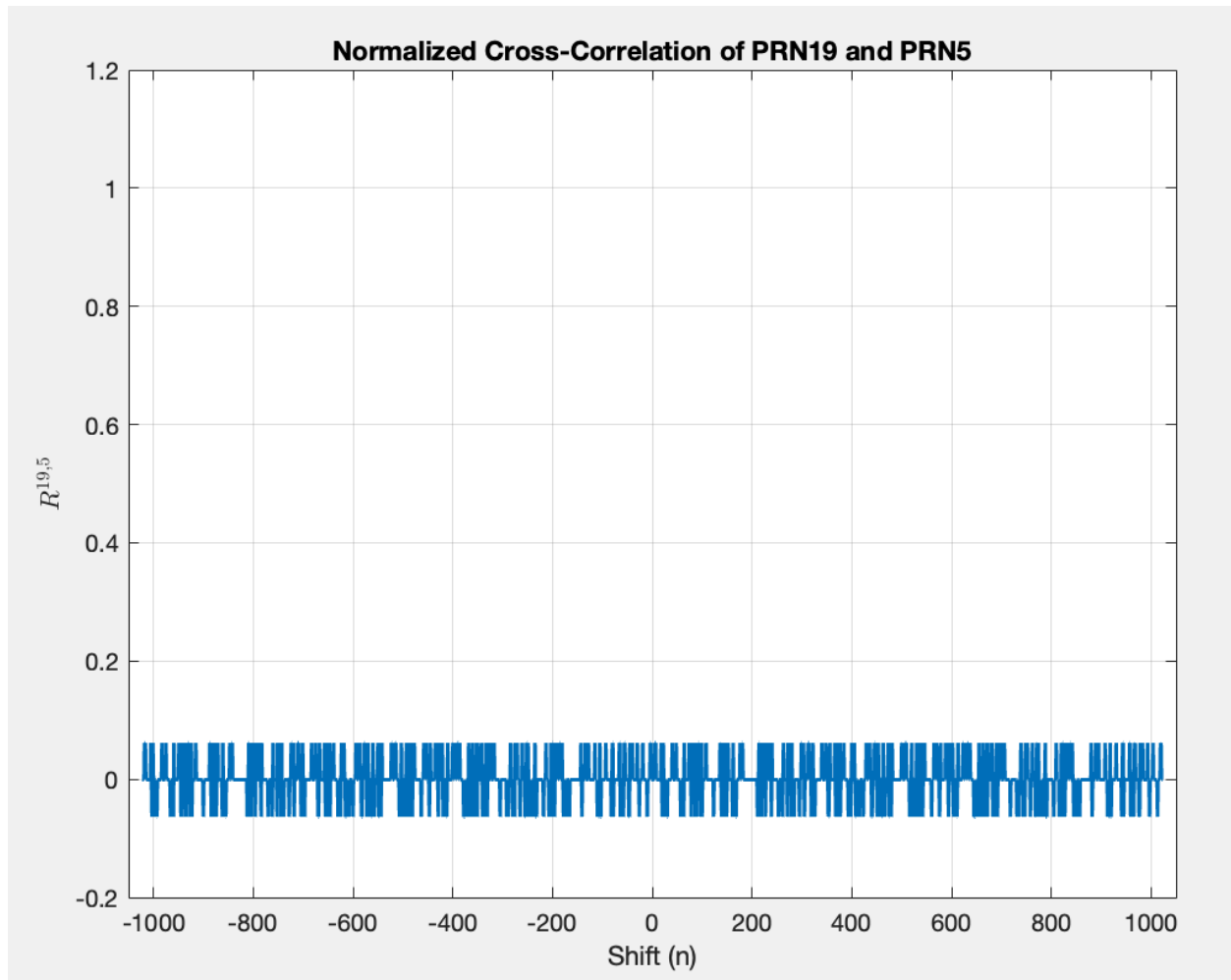
Here's the cross-correlation function for PRN19 and PRN25:



Compared to part (a), there are no peaks of maximum correlation between PRN19 and PRN25. This suggests that PRN19 and PRN25 would not interfere with each other.

Problem 4d

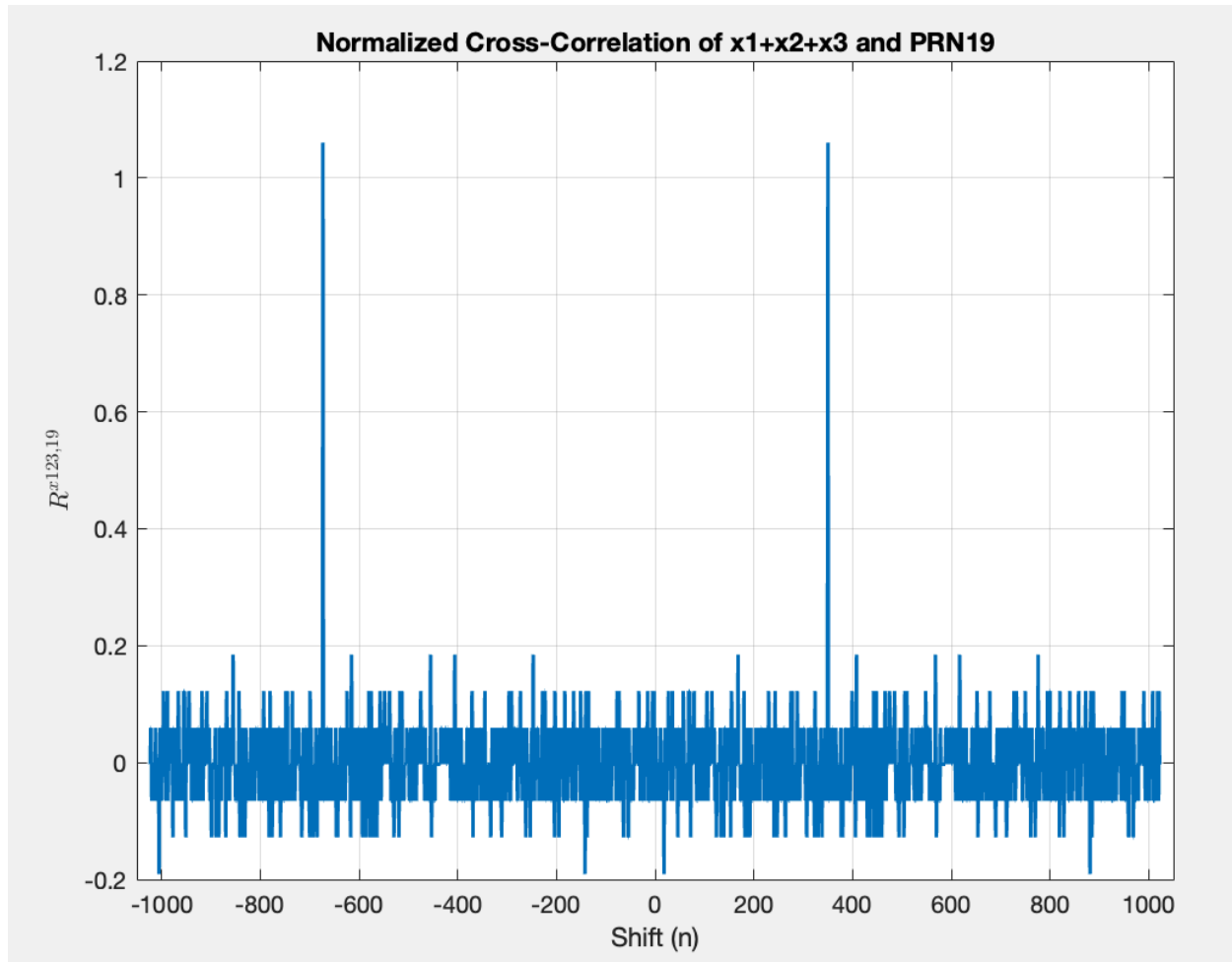
Here's the cross-correlation function for PRN19 and PRN5:



Similar to part (c), there are no peaks of maximum correlation between PRN19 and PRN5. This suggests that PRN19 and PRN5 would not interfere with each other.

Problem 4e

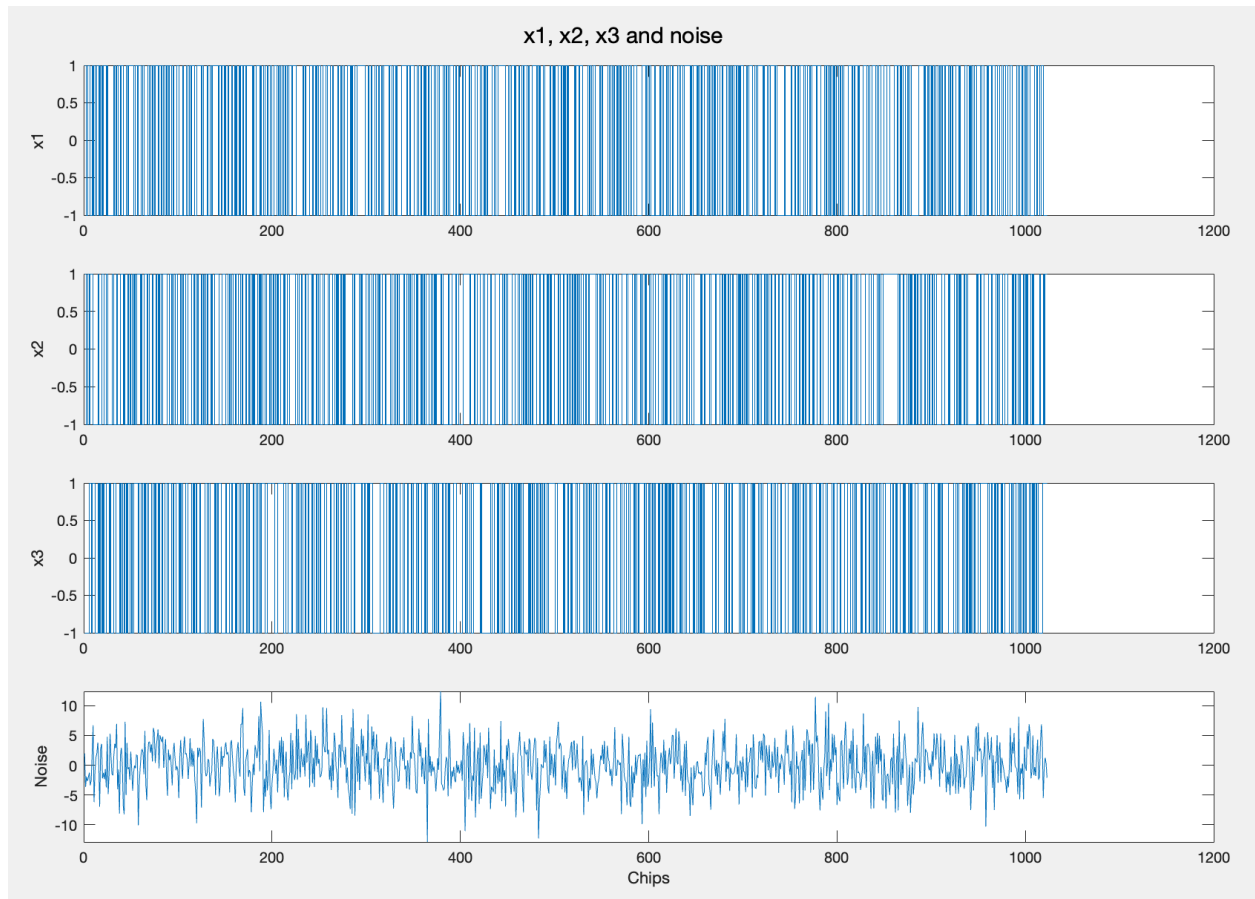
Here's the cross-correlation function for sum of x1 (PRN19 delayed by 350), x2 (PRN25 delayed by 905) and x3 (PRN5 delayed by 75) and PRN19:



Following the logic in part (a), (b), (c), (d), it is evident that when similar PRNs are correlated with each other, the peaks are shifted by the delay (to the right) and when different PRNs are correlated with each other, they do not interfere (end up in the noise). So, the peaks of correlation between PRN19 and PRN19 delayed by 350 chips is evident in this plot (peaks at $n = 350$ and -673). The other PRNs do not interfere.

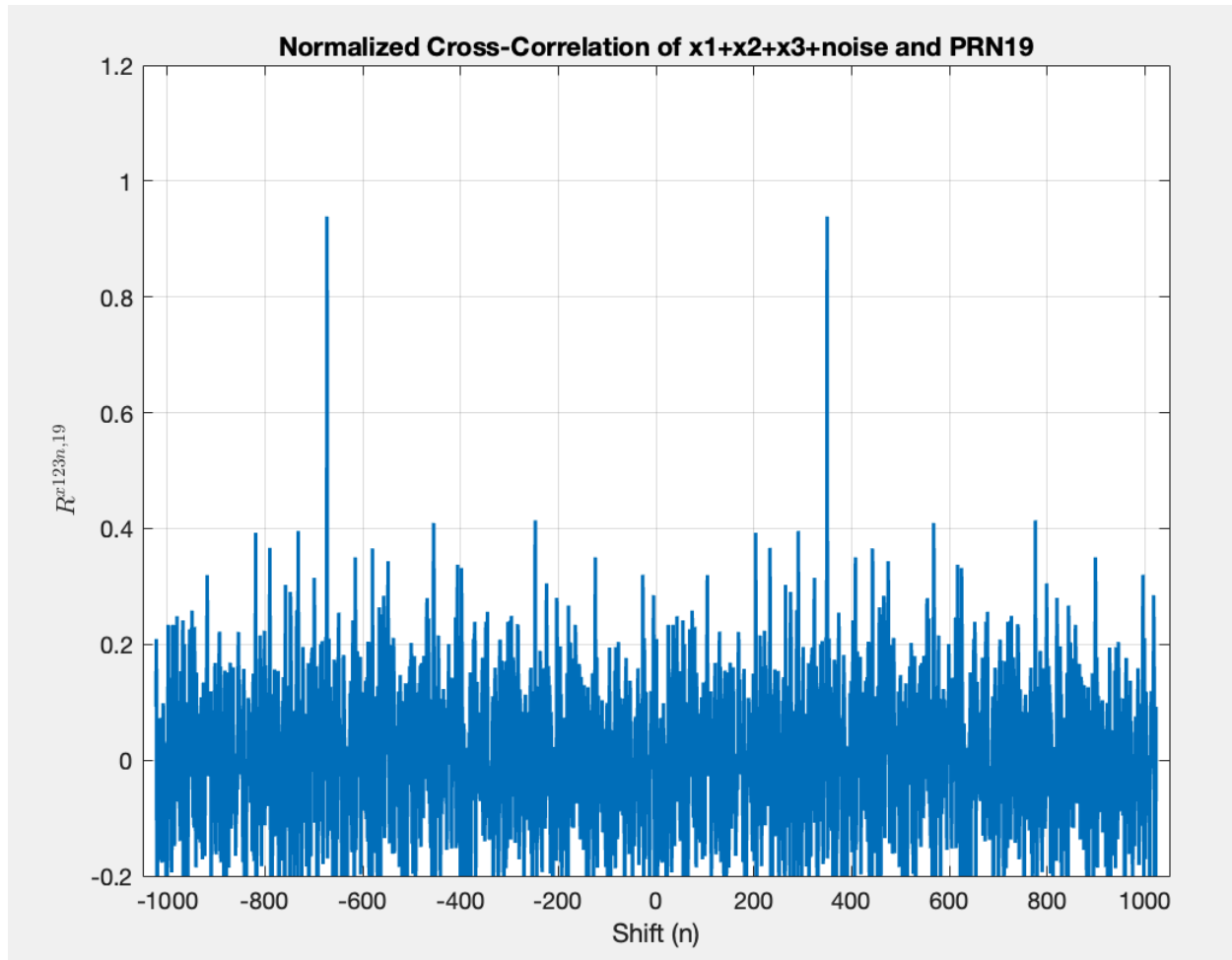
Problem 4f

Here's the plot for x_1 , x_2 , x_3 , and noise:



Problem 4g

Here's the cross-correlation function for sum of x1 (PRN19 delayed by 350), x2 (PRN25 delayed by 905), x3 (PRN5 delayed by 75), and noise (0 mean and standard deviation of 4) and PRN19:



As seen in part (e), PRN19 was visible when interfered with other PRNs, but in this case, it also seems to be distinct from noise along with other PRNs.

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```
clear; clc; close all;
```

ASEN 5090 - HW 1

Jash Bhalavat 08/25/2026

```
% Constants

% Speed of light in a vacuum
c = 2.99792458e8; % m/s, ME Table 4.1
```

Problem 1a

```
rho_PL1 = 550; % m
rho_PL2 = 500; % m

delta_r = (rho_PL1 - rho_PL2) / (2*c); % sec
R_r_PL1 = rho_PL1 - c*delta_r; % m
R_r_PL2 = 1000 - R_r_PL1; % m

n = 500;
```

Problem 2

```
v0 = 50; % m/s
h0 = 100; % m
x0 = 250; % m

xA = linspace(-x0, x0, n);

h0_err = 10/100;
```

```

h0_neg = h0 - h0_err;
h0_pos = h0 + h0_err;

for i = 1:n
    rho(i) = sqrt(h0^2 + xA(i)^2); % m
    rho_neg(i) = sqrt(h0_neg^2 + xA(i)^2); % m
    rho_pos(i) = sqrt(h0_pos^2 + xA(i)^2); % m
    rho_dot(i) = v0*xA(i) / rho(i); % m/s
    rho_dot_neg(i) = v0*xA(i) / rho_neg(i); % m/s
    rho_dot_pos(i) = v0*xA(i) / rho_pos(i); % m/s
    % z(i) = asin(xA(i) / rho(i)); %
    z(i) = atan2(xA(i), h0);
    z_neg(i) = atan2(xA(i), h0_neg);
    z_pos(i) = atan2(xA(i), h0_pos);
end

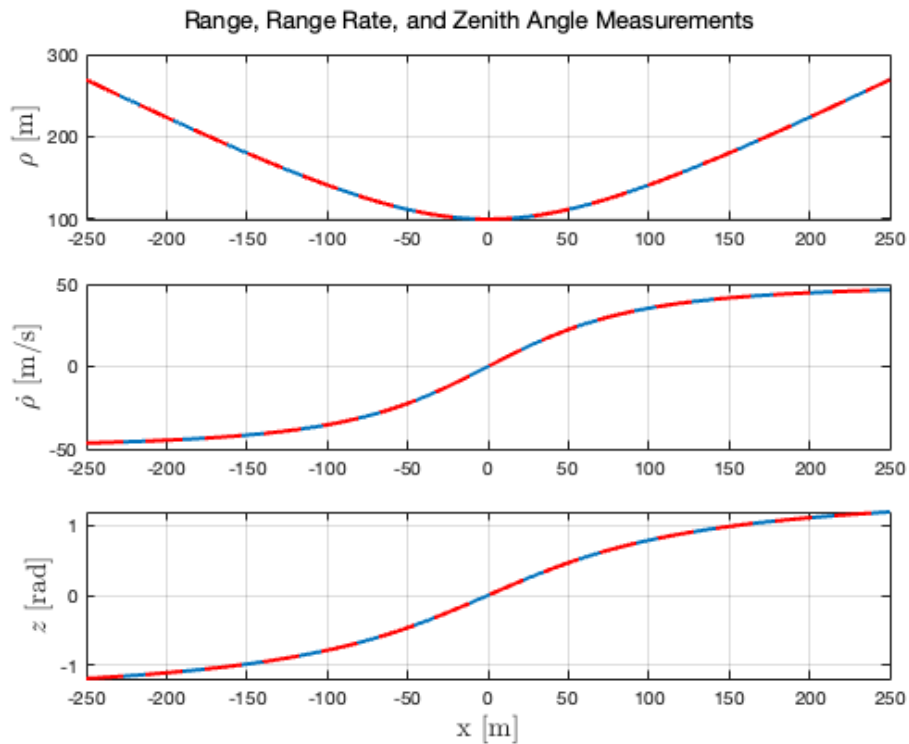
% Plot
figure(1)
subplot(3,1,1)
plot(xA, rho, 'LineWidth', 2)
hold on
plot(xA, rho_neg, '--', 'Color', 'red', 'LineWidth', 2)
plot(xA, rho_pos, '--', 'Color', 'red', 'LineWidth', 2)
hold off
ylabel('$\rho$ [m]', 'Interpreter', 'Latex', 'FontSize', 15)
grid on

subplot(3,1,2)
plot(xA, rho_dot, 'LineWidth', 2)
hold on
plot(xA, rho_dot_neg, '--', 'Color', 'red', 'LineWidth', 2)
plot(xA, rho_dot_pos, '--', 'Color', 'red', 'LineWidth', 2)
hold off
ylabel('$\dot{\rho}$ [m/s]', 'Interpreter', 'Latex', 'FontSize', 15)
grid on

subplot(3,1,3)
plot(xA, z, 'LineWidth', 2)
hold on
plot(xA, z_neg, '--', 'Color', 'red', 'LineWidth', 2)
plot(xA, z_pos, '--', 'Color', 'red', 'LineWidth', 2)
hold off
ylabel('$z$ [rad]', 'Interpreter', 'Latex', 'FontSize', 15)
xlabel('x [m]', 'Interpreter', 'latex', 'FontSize', 15)
grid on

sgtitle("Range, Range Rate, and Zenith Angle Measurements")

```



Problem 3a

```
% For PRN 19, S1 = 3, S2 = 6. Source = https://archive.gps.gov/technical/icwg/IS-GPS-200D.pdf
S1 = 3;
S2 = 6;

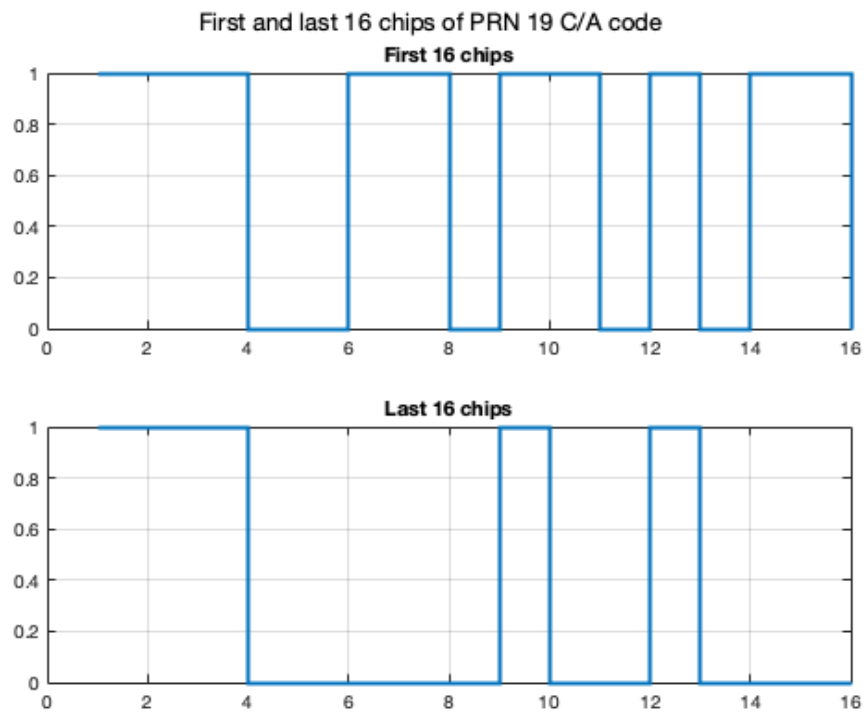
ca_code_prn19_1023 = ca_code(1023, S1, S2);

prn19_first16_hex = dec2hex(bin2dec(num2str(ca_code_prn19_1023(1:16))));
prn19_last16_hex = dec2hex(bin2dec(num2str(ca_code_prn19_1023(end-15:end))));

figure(1)
subplot(2,1,1)
stairs(ca_code_prn19_1023(1:16), 'linewidth', 2)
title("First 16 chips")
grid on

subplot(2,1,2)
stairs(ca_code_prn19_1023(end-15:end), 'linewidth', 2)
title("Last 16 chips")
grid on

sgtitle("First and last 16 chips of PRN 19 C/A code")
```

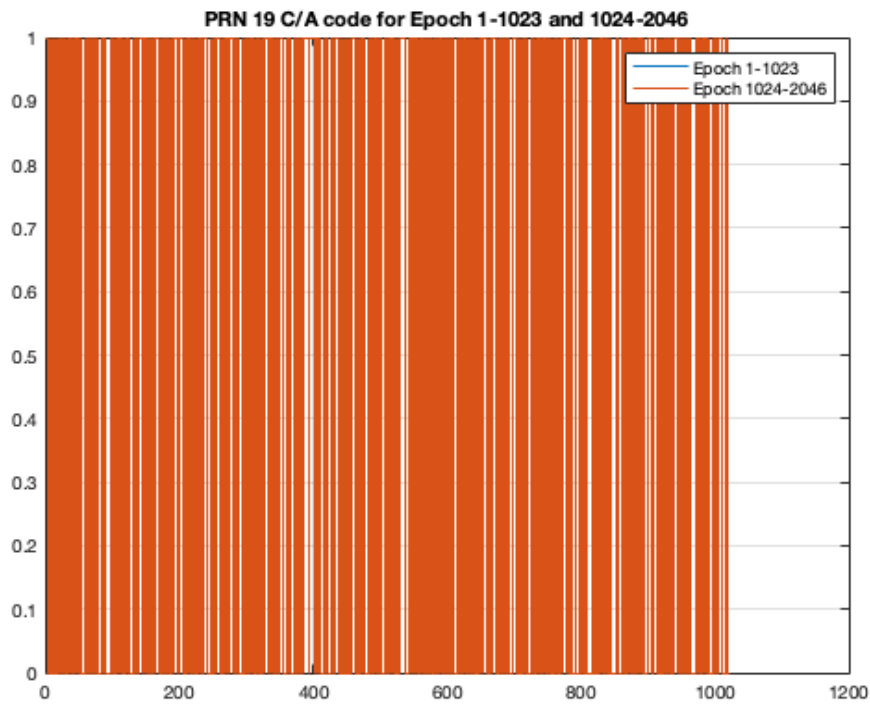


Problem 3b

```
ca_code_prn19_2046 = ca_code(2046, S1, S2);

figure(2)
stairs(ca_code_prn19_2046(1:1023), 'linewidth', 1)
hold on
stairs(ca_code_prn19_2046(1024:2046), 'linewidth', 1)
hold off
legend("Epoch 1-1023", "Epoch 1024-2046")
grid on

title("PRN 19 C/A code for Epoch 1-1023 and 1024-2046")
```



Problem 3c

```
% For PRN 25, S1 = 5, S2 = 7. Source = https://archive.gps.gov/technical/icwg/IS-GPS-200D.pdf
S1 = 5;
S2 = 7;

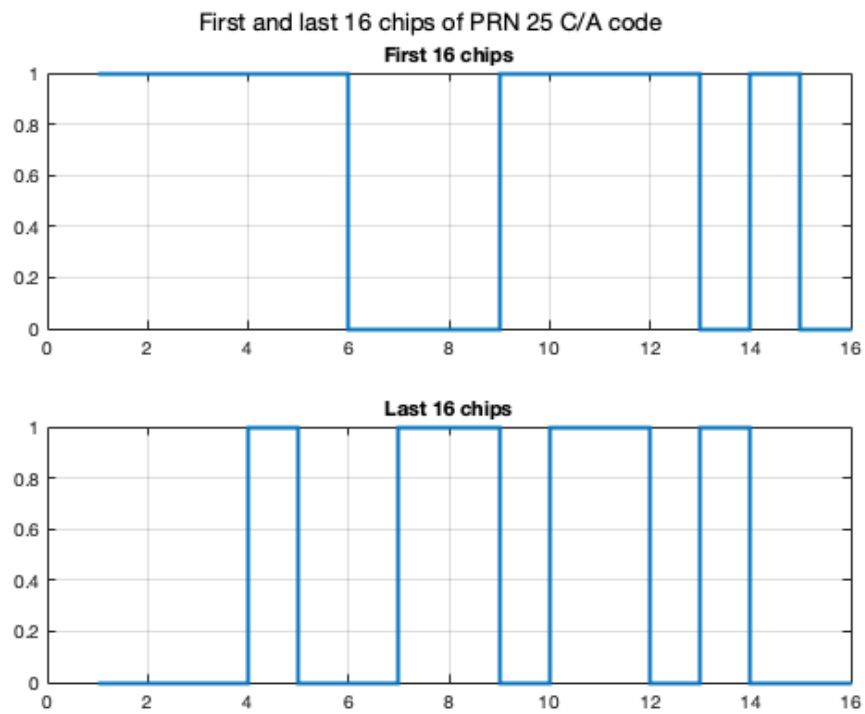
ca_code_prn25_1023 = ca_code(1023, S1, S2);

prn25_first16_hex = dec2hex(bin2dec(num2str(ca_code_prn25_1023(1:16))));
prn25_last16_hex = dec2hex(bin2dec(num2str(ca_code_prn25_1023(end-15:end))));

figure(3)
subplot(2,1,1)
stairs(ca_code_prn25_1023(1:16), 'linewidth', 2)
title("First 16 chips")
grid on

subplot(2,1,2)
stairs(ca_code_prn25_1023(end-15:end), 'linewidth', 2)
title("Last 16 chips")
grid on

sgtitle("First and last 16 chips of PRN 25 C/A code")
```



Problem 3d

```
% For PRN 5, S1 = 1, S2 = 9. Source = https://archive.gps.gov/technical/icwg/IS-GPS-200D.pdf
S1 = 1;
S2 = 9;

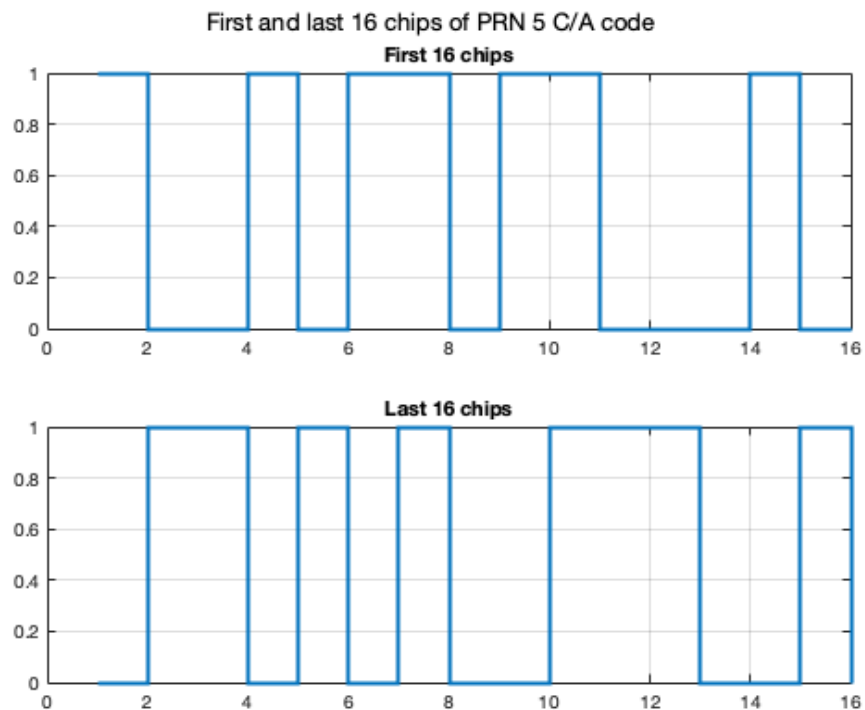
ca_code_prn5_1023 = ca_code(1023, S1, S2);

prn5_first16_hex = dec2hex(bin2dec(num2str(ca_code_prn5_1023(1:16))));
prn5_last16_hex = dec2hex(bin2dec(num2str(ca_code_prn5_1023(end-15:end))));

figure(4)
subplot(2,1,1)
stairs(ca_code_prn5_1023(1:16), 'linewidth', 2)
title("First 16 chips")
grid on

subplot(2,1,2)
stairs(ca_code_prn5_1023(end-15:end), 'linewidth', 2)
title("Last 16 chips")
grid on

sgtitle("First and last 16 chips of PRN 5 C/A code")
```



Problem 4a

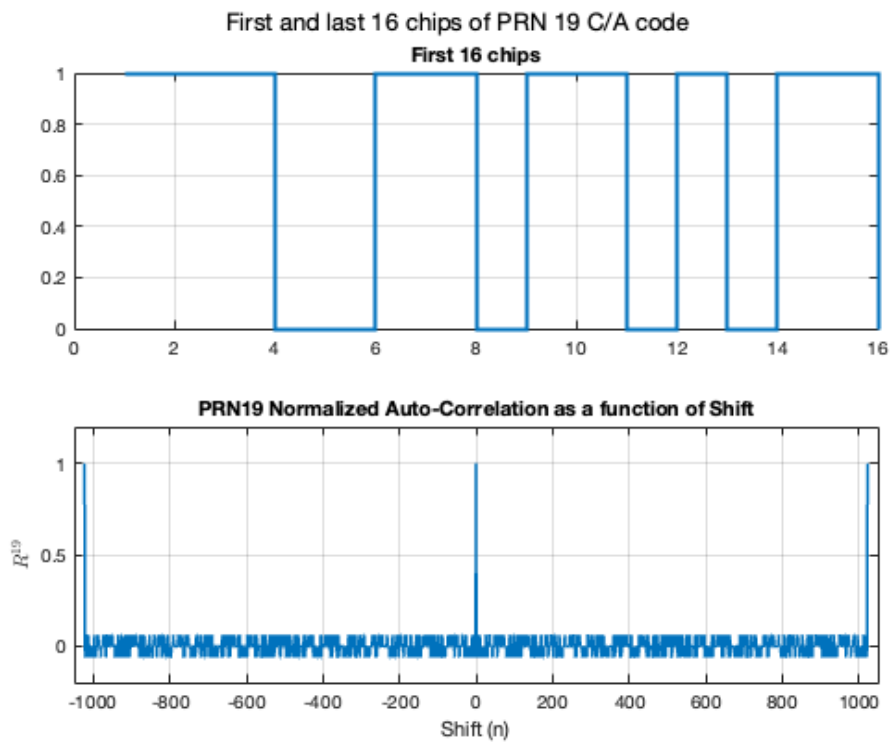
```
% For PRN 19, S1 = 3, S2 = 6. Source = https://archive.gps.gov/technical/icwg/IS-GPS-200D.pdf
S1 = 3;
S2 = 6;

ca_code_prn19_1023 = ca_code(1023, S1, S2);
ca_problem2_prn19_1023 = (ca_code_prn19_1023==0)*(1) + (ca_code_prn19_1023==1)*(-1);

n = linspace(-1023, 1023, 2047);

for i = 1:length(n)
    R19(i) = auto_correlation_norm(n(i), ca_problem2_prn19_1023);
end

figure(1)
plot(n, R19, 'LineWidth', 1.5)
xlim([-1050, 1050])
xlabel("Shift (n)")
ylabel("$R^{19}$", 'Interpreter', 'latex')
title("PRN19 Normalized Auto-Correlation as a function of Shift")
grid on
ylim([-0.2, 1.2])
```

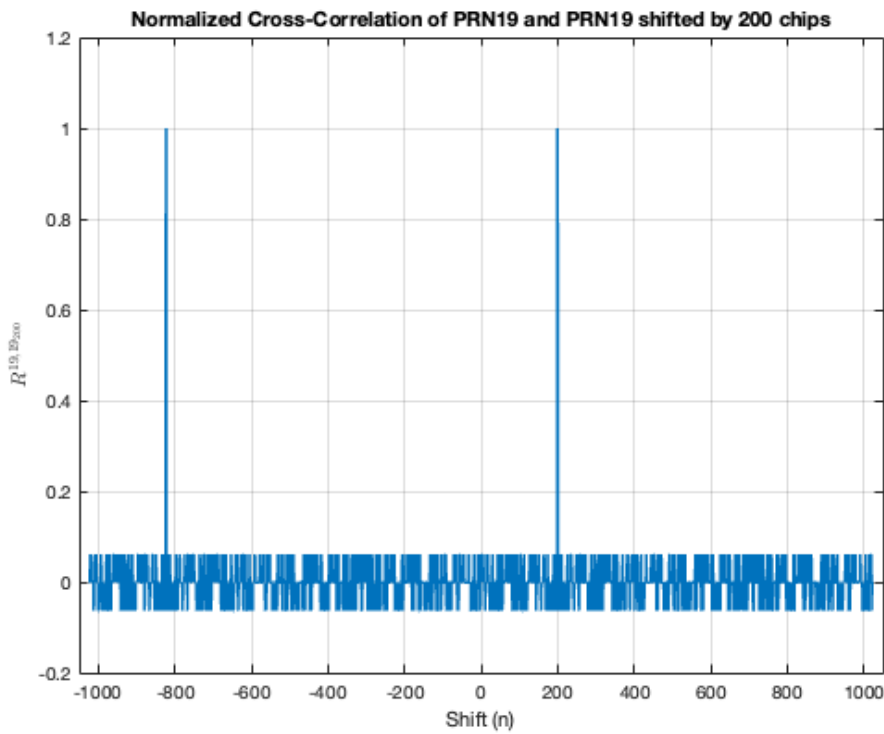


Problem 4b

```
ca_problem2_prn19_1023_shifted_by_200 = circshift(ca_problem2_prn19_1023, [0, 200]);

for i = 1:length(n)
    R19_19200(i) = cross_correlation_norm(n(i), ca_problem2_prn19_1023_shifted_by_200, ca_problem2_prn19_1023);
end

figure(2)
plot(n, R19_19200, 'LineWidth', 1.5)
xlim([-1050, 1050])
xlabel("Shift (n)")
ylabel("$R^{19,19}_{200}$", 'Interpreter', 'latex')
title("Normalized Cross-Correlation of PRN19 and PRN19 shifted by 200 chips")
grid on
ylim([-0.2, 1.2])
```



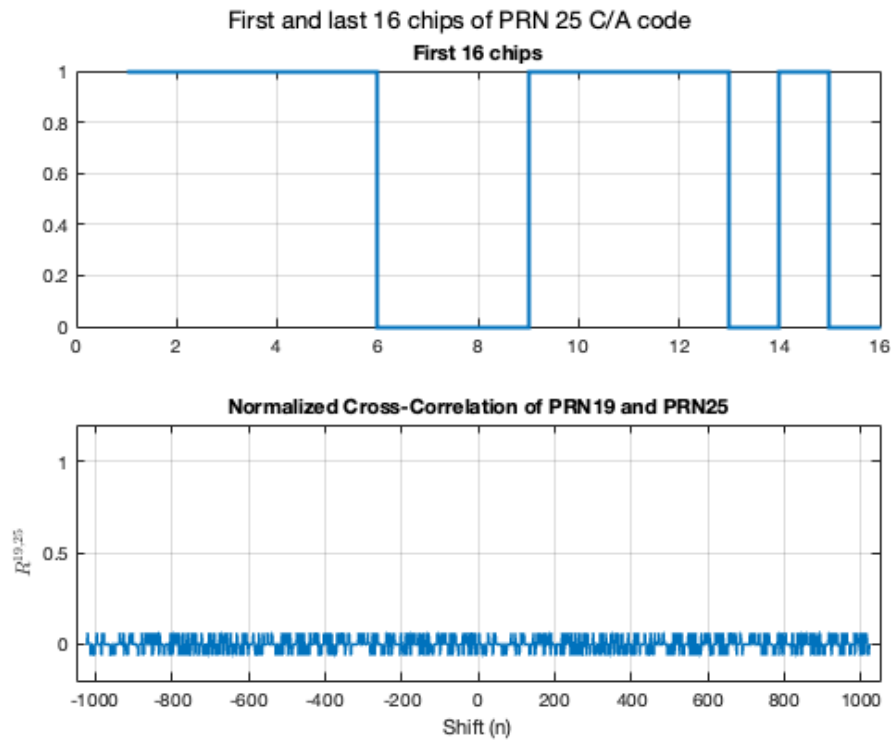
Problem 4c

```
% For PRN 25, S1 = 5, S2 = 7. Source = https://archive.gps.gov/technical/icwg/IS-GPS-200D.pdf
S1 = 5;
S2 = 7;

ca_code_prn25_1023 = ca_code(1023, S1, S2);
ca_problem2_prn25_1023 = (ca_code_prn25_1023==0)*(1) + (ca_code_prn25_1023==1)*(-1);

for i = 1:length(n)
    R19_25(i) = cross_correlation_norm(n(i), ca_problem2_prn25_1023, ca_problem2_prn19_1023);
end

figure(3)
plot(n, R19_25, 'LineWidth', 1.5)
xlim([-1050, 1050])
xlabel("Shift (n)")
ylabel("$R^{19,25}$", 'Interpreter', 'latex')
title("Normalized Cross-Correlation of PRN19 and PRN25")
grid on
ylim([-0.2, 1.2])
```



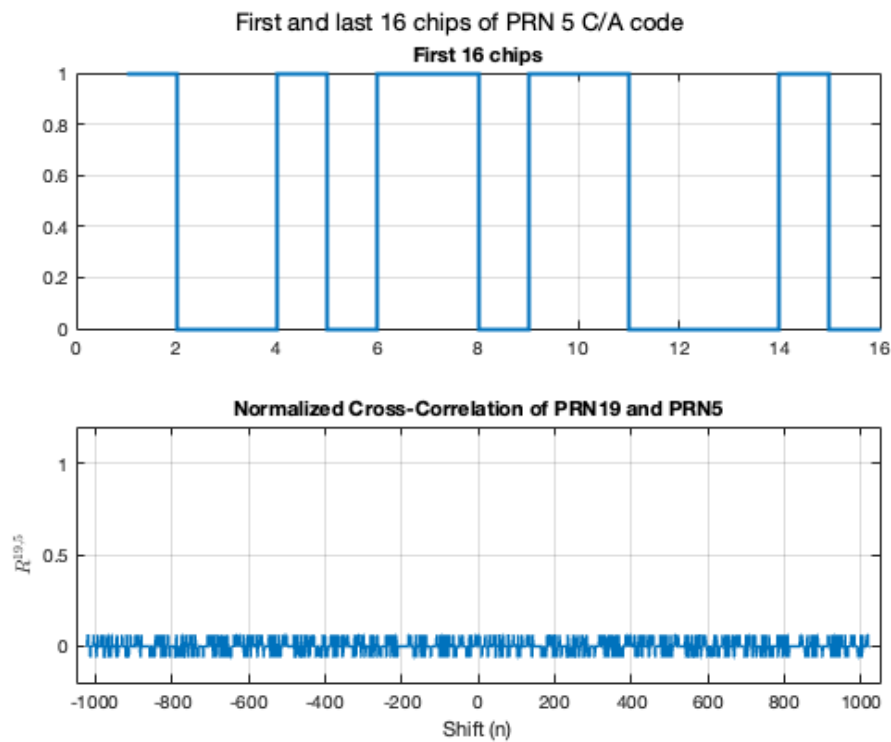
Problem 4d

```
% For PRN 5, S1 = 1, S2 = 9. Source = https://archive.gps.gov/technical/icwg/IS-GPS-200D.pdf
S1 = 1;
S2 = 9;

ca_code_prn5_1023 = ca_code(1023, S1, S2);
ca_problem2_prn5_1023 = (ca_code_prn5_1023==0)*(1) + (ca_code_prn5_1023==1)*(-1);

for i = 1:length(n)
    R19_5(i) = cross_correlation_norm(n(i), ca_problem2_prn5_1023, ca_problem2_prn19_1023);
end

figure(4)
plot(n, R19_5, 'LineWidth', 1.5)
xlim([-1050, 1050])
xlabel("Shift (n)")
ylabel("$R^{19,5}$", 'Interpreter', 'latex')
title("Normalized Cross-Correlation of PRN19 and PRN5")
grid on
ylim([-0.2, 1.2])
```

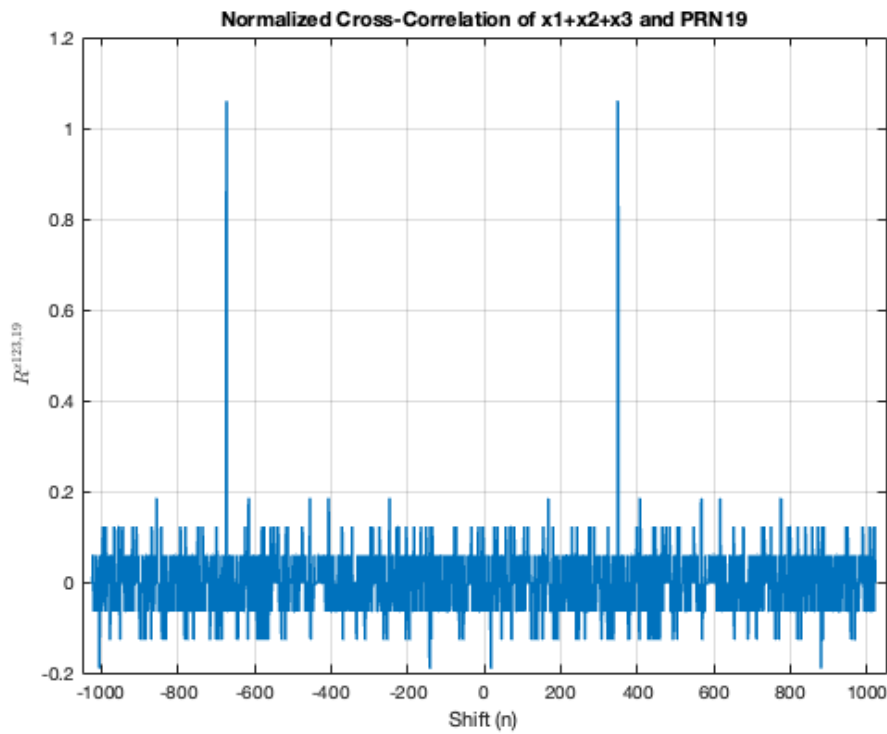


Problem 4e

```
x1 = circshift(ca_problem2_prn19_1023, 350);
x2 = circshift(ca_problem2_prn25_1023, 905);
x3 = circshift(ca_problem2_prn5_1023, 75);
x123 = x1 + x2 + x3;

for i = 1:length(n)
    Rx123_19(i) = cross_correlation_norm(n(i), x123, ca_problem2_prn19_1023);
end

figure(5)
plot(n, Rx123_19, 'LineWidth', 1.5)
xlim([-1050, 1050])
xlabel("Shift (n)")
ylabel("$R^{\{x123,19\}}$", 'Interpreter', 'latex')
title("Normalized Cross-Correlation of x1+x2+x3 and PRN19")
grid on
ylim([-0.2, 1.2])
```



Problem 4f

```
noise = 4*randn(1,1023);

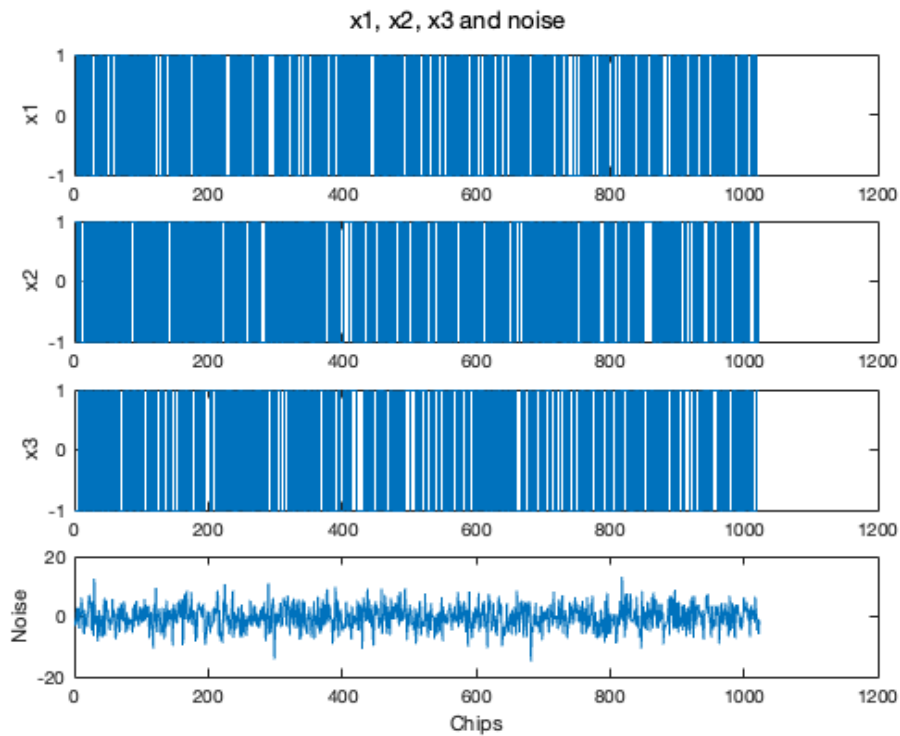
figure(6)
sgtitle("x1, x2, x3 and noise")

subplot(4, 1, 1)
stairs(x1)
ylabel("x1")

subplot(4, 1, 2)
stairs(x2)
ylabel("x2")

subplot(4, 1, 3)
stairs(x3)
ylabel("x3")

subplot(4, 1, 4)
plot(noise)
ylabel("Noise")
xlabel("Chips")
```

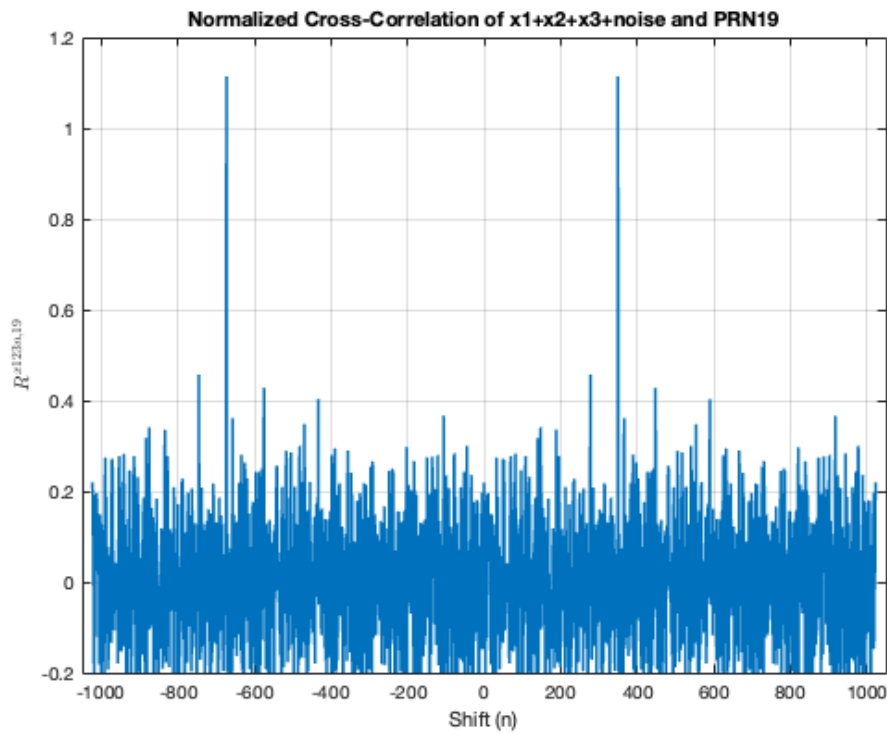


Problem 4g

```
x123n = x123 + noise;

for i = 1:length(n)
    Rx123n_19(i) = cross_correlation_norm(n(i), x123n, ca_problem2_prn19_1023);
end

figure(7)
plot(n, Rx123n_19, 'LineWidth', 1.5)
xlim([-1050, 1050])
xlabel("Shift (n)")
ylabel("$R^{\{x123n,19\}}$", 'Interpreter', 'latex')
title("Normalized Cross-Correlation of x1+x2+x3+noise and PRN19")
grid on
ylim([-0.2, 1.2])
```



Functions

```
function out = auto_correlation_norm(n, x_k)
    % Normalized auto correlation function
    % n - shift
    % x_k - PRN associated with satellite k

    out = sum(x_k .* circshift(x_k, n))/1023;

end

function out = cross_correlation_norm(n, x_k, x_l)
    % Normalized cross correlation function
    % n - shift
    % x_k - PRN associated with satellite k
    % x_l - PRN associated with satellite l

    out = sum(x_k .* circshift(x_l, [0, n]))/1023;

end
```